

ASG AIRLINES – END-TO-END DATA ENGINEERING AND ANALYTICS PIPELINE

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1. Executive Summary

ASG Airlines operates as a nationwide airline and generates large volumes of operational data through flight bookings, scheduling systems, airport operations, passenger records, and payment transactions.

The objective of this project is to design and implement an end-to-end data engineering pipeline that ingests airline operational data, performs data cleaning and validation, standardizes inconsistent values, handles missing information, calculates flight durations, identifies overnight flights, protects passenger Personally Identifiable Information (PII), and produces an analytical dataset for business intelligence.

The solution was implemented using Python and Pandas for local data engineering and transformation. Power BI was used to develop an interactive analytical dashboard.

The complete workflow follows:

Raw Data → Ingestion → Data Quality & Cleaning → Transformation → Analytical Data Layer → Power BI

The solution also includes data quality checks, PII masking, referential integrity validation, KPI generation, and documentation of assumptions and transformation rules.

2. Project Objectives

The major objectives of the project are:

1. Ingest airline operational datasets from the provided Excel workbook.
2. Inspect and understand the structure and quality of the data.
3. Identify missing values and inconsistent data.
4. Validate flight identifiers.
5. Standardize text and categorical values.
6. Convert inconsistent timestamp values into a standardized datetime format.
7. Calculate flight duration from departure and arrival timestamps.
8. Correctly identify overnight flights.
9. Generate useful analytical features.
10. Validate relationships between flights, bookings, passengers, and payments.
11. Protect passenger PII information.
12. Create cleaned datasets suitable for visualization.
13. Create analytical fact and dimension datasets.
14. Calculate operational KPIs.
15. Build an interactive Power BI dashboard.
16. Design a scalable architecture that can later be migrated to Azure.
17. Document the complete workflow, assumptions, cleaning strategy, transformations, and data model.

3. Dataset Documentation

The supplied Excel workbook contains four major datasets:

Dataset	Rows	Columns	Purpose
Flights	1,020	7	Flight operational information
Bookings	1,000	9	Passenger flight booking information
Passengers	1,039	9	Passenger information
Payments	1,000	4	Payment transaction information

3.1 Flights Dataset

The `flights` table contains information about individual flights.

Columns

Column	Description
<code>flight_id</code>	Unique flight identifier
<code>airline</code>	Airline operating the flight
<code>source</code>	Departure airport
<code>destination</code>	Arrival airport
<code>departure_time</code>	Flight departure timestamp
<code>arrival_time</code>	Flight arrival timestamp
<code>duration</code>	Recorded flight duration

Example:

<code>flight_id</code>	<code>airline</code>	<code>source</code>	<code>destination</code>
SJ010	SpiceJet	CCU	MAA
AI155	Air India	BOM	CCU
UK094	Vistara	BOM	CCU

4. Bookings Dataset

The bookings dataset contains booking and reservation information.

Columns

Column	Description
booking_id	Unique booking identifier
passenger_id	Passenger reference
flight_id	Flight reference
booking_date	Date of booking
status	Booking status
passport_number	Passenger passport information
seat_number	Allocated seat
emergency_contact_name	Emergency contact
emergency_contact_phone	Emergency contact phone

Because this dataset contains sensitive information, passport and emergency contact information should not be included in publicly exposed analytical datasets.

5. Passengers Dataset

The passengers dataset contains passenger-level information.

Columns

Column	Description
passenger_id	Passenger identifier
first_name	Passenger first name
last_name	Passenger last name
age	Passenger age
gender	Passenger gender
email	Passenger email
phone	Passenger phone
aadhaar_id	Aadhaar identifier
date_of_birth	Passenger date of birth

This table contains significant PII and therefore requires additional protection.

6. Payments Dataset

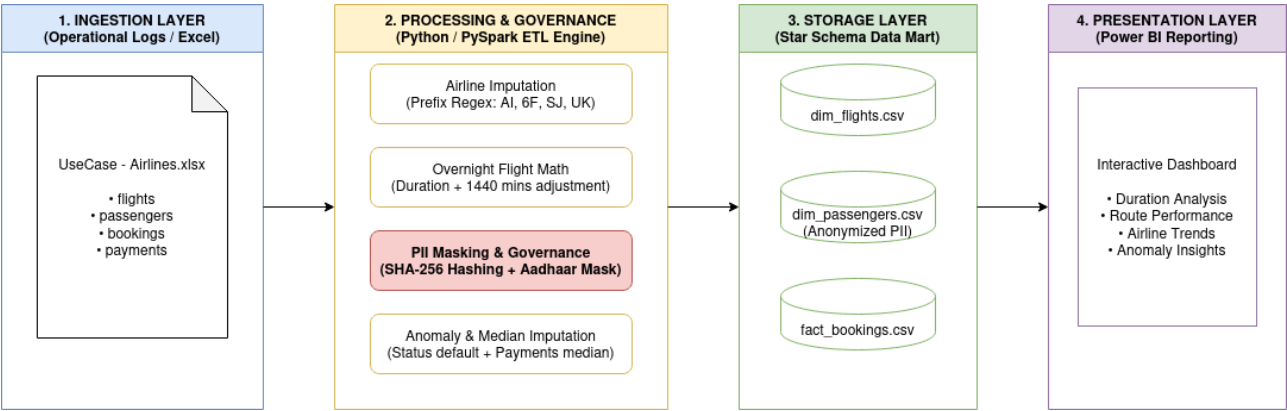
The payment dataset contains transaction information.

Columns

Column	Description
payment_id	Unique payment identifier
booking_id	Related booking
amount	Payment amount
payment_method	Payment method used

7. Proposed System Architecture

The proposed architecture consists of multiple logical layers.



Architecture Components:

7.1 Raw Data Layer

The original Excel workbook is treated as the raw source.

The raw layer preserves the original data without modifying the source records.

7.2 Data Ingestion Layer

Python and Pandas are used to read the Excel workbook and load each sheet independently.

The four datasets are loaded into separate DataFrames:

- Flights
- Bookings
- Passengers
- Payments

7.3 Data Quality and Cleaning Layer

This layer performs:

- Missing-value analysis
- Duplicate detection
- Data type validation
- Flight ID validation
- Timestamp validation
- Text standardization
- Referential integrity checks
- PII protection

7.4 Transformation Layer

The transformation layer generates:

- Calculated duration
- Duration in minutes
- Duration in hours
- Overnight indicator
- Route
- Departure date
- Arrival date
- Departure hour
- Arrival hour
- Departure day

7.5 Analytical Data Layer

The transformed data is organized into fact and dimension datasets.

Examples:

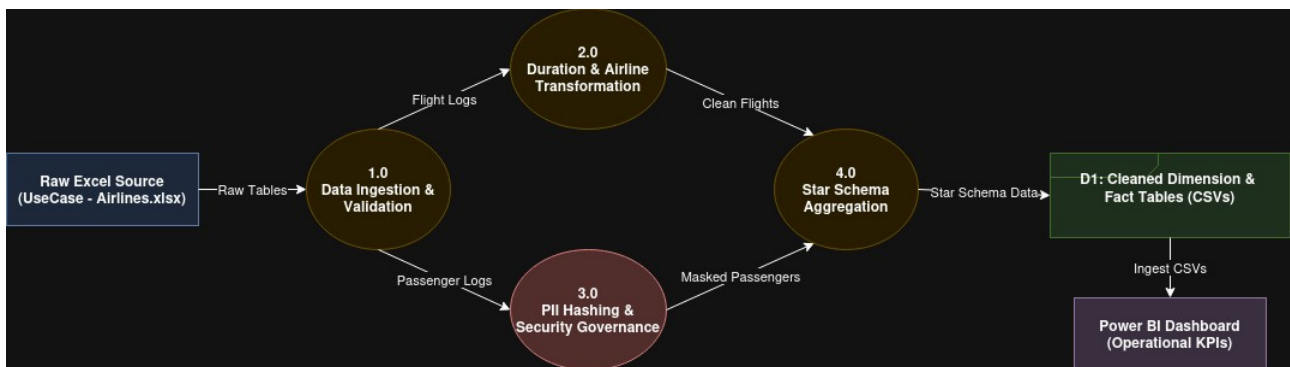
- FACT_FLIGHTS
- FACT_BOOKINGS
- FACT_PAYMENTS
- DIM_AIRLINE
- DIM_ROUTE
- DIM_PASSENGER

7.6 Reporting Layer

Power BI connects to the analytical datasets and provides interactive reports, KPI cards, charts, slicers, and operational insights.

8. Data Flow Diagram

The complete data flow is:



9. Data Cleaning Strategy

The raw data contains intentional quality issues. A structured cleaning strategy was therefore applied.

9.1 Missing Value Detection

Missing values were identified for each dataset and column.

Observed missing values include:

Flights

- Airline: 41 missing records

Bookings

- Status: 45 missing records

Passengers

- Last name: 10 missing records

Payments

- Amount: 48 missing records

Missing values were not blindly removed. Each field was considered according to its business meaning.

10. Flight ID Validation

Flight IDs were standardized by:

1. Removing leading and trailing spaces.
2. Converting identifiers to uppercase.
3. Checking the identifier against a defined format.

Example validation pattern:

```
^[A-Z0-9]{5}$
```

The validation identifies IDs that do not conform to the expected five-character alphanumeric format.

Invalid records are flagged rather than automatically deleted.

This allows the data quality layer to preserve the original information while identifying problematic records for further investigation.

11. Text Standardization

Text fields such as airline, source airport, and destination airport were standardized.

The following operations were applied:

- Leading/trailing whitespace removal
- Conversion to uppercase
- Consistent representation of categorical values

For example:

```
"Air India"  
"air india"  
"AIR INDIA"
```

can be standardized to:

```
AIR INDIA
```

Missing airline values are represented using an explicit UNKNOWN category where appropriate.

12. Timestamp Standardization

- Departure and arrival timestamps may contain different formats.
- All timestamp values are converted into a standard datetime representation.
- Invalid timestamp values are converted into missing datetime values and flagged for data quality review.
- The standardized timestamps are then used for duration calculation and temporal analysis.

13. Flight Duration Calculation

Flight duration is calculated using:

`Flight Duration = Arrival Time - Departure Time`

The calculated duration is converted into minutes.

For example:

`Departure = 23:38`

`Arrival = 02:32 next day`

The calculation must consider the date component.

Therefore, the correct duration is approximately:

`2 hours 54 minutes`

rather than treating the arrival time as occurring before departure.

14. Overnight Flight Handling

Overnight flights are flights where the arrival date is later than the departure date.

The following business rule is used:

`Arrival Date > Departure Date`

If this condition is true:

`is_overnight = TRUE`

Otherwise:

`is_overnight = FALSE`

This ensures that flights crossing midnight are handled correctly.

15. Duration Validation

The original recorded duration is compared against the duration calculated from timestamps.

The following values are generated:

- Original duration
- Calculated duration
- Duration difference

The difference is calculated as:

Duration Difference =
Calculated Duration - Original Duration

This helps identify inconsistent duration records.

16. Route Creation

A standardized route field is created using:

Source → Destination

For example:

BOM → CCU
CCU → MAA

This route field is useful for:

- Route traffic analysis
- Average duration analysis
- Route ranking
- Power BI visualizations

17. Time-Based Feature Engineering

Additional time features are generated from the flight timestamps.

These include:

Feature	Description
departure_date	Date of departure
arrival_date	Date of arrival
departure_hour	Hour of departure
arrival_hour	Hour of arrival
departure_day	Day of the week
is_overnight	Overnight indicator

These features allow operational patterns to be analyzed.

18. PII Protection

Passenger information contains sensitive Personally Identifiable Information.

Sensitive fields include:

- First name
- Last name
- Email
- Phone
- Aadhaar
- Date of birth
- Passport number
- Emergency contact details

These fields should not be exposed in the public analytical dataset or GitHub repository.

18.1 Passenger Identifier Hashing

A SHA-256 hash can be generated for passenger identifiers.

Instead of exposing the original passenger identifier:

P10001

the analytical layer can use a hashed passenger key.

This allows records to be linked while reducing direct exposure of the original identifier.

18.2 Aadhaar Masking

If Aadhaar information is required for controlled internal processing, it can be masked so that only the final four characters are visible.

Example:

*****1234

However, Aadhaar should preferably be excluded entirely from analytical and public datasets unless there is a justified business requirement.

18.3 Public Repository Rule

The following information should **not** be uploaded to a public GitHub repository:

- Raw passenger data
- Aadhaar numbers
- Passport numbers
- Personal phone numbers

- Personal email addresses
- Emergency contact information

Only masked or anonymized analytical data should be included.

19. Data Quality Flags

A data quality flag is generated for flight records.

Possible statuses include:

VALID
INVALID_FLIGHT_ID
INVALID_TIMESTAMP
NEGATIVE_DURATION
MISSING_DURATION

The flag allows Power BI to display the overall quality of the dataset.

A better implementation can maintain multiple quality indicators so that a record containing more than one issue does not lose information about earlier detected issues.

20. Referential Integrity

Relationships between datasets are validated.

Booking → Flight

Every booking's `flight_id` should exist in the flights dataset.

Booking → Passenger

Every booking's `passenger_id` should exist in the passenger dataset.

Payment → Booking

Every payment's `booking_id` should exist in the booking dataset.

This ensures that relationships between datasets remain reliable.

21. Analytical Data Model

The analytical model follows a fact-and-dimension structure.

Dimension Tables

DIM_AIRLINE

Column
<code>airline_key</code>
<code>airline</code>

DIM_ROUTE

Column
route_key
source
destination
route

DIM_PASSENGER

Column
passenger_key
age
gender

Sensitive passenger fields are excluded from the analytical layer.

22. Fact Tables

FACT_FLIGHTS

Column
flight_id
airline
source
destination
route
departure_time
arrival_time
departure_date
arrival_date
departure_hour
arrival_hour
departure_day
duration_minutes
duration_hours
is_overnight
data_quality_flag

FACT_BOOKINGS

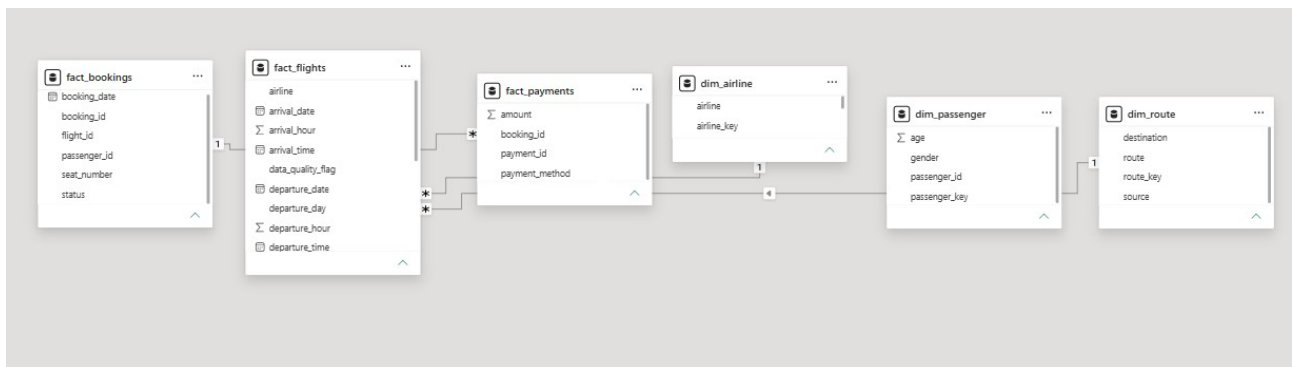
Column
booking_id
passenger_id / passenger_key
flight_id
booking_date
status
seat_number

FACT_PAYMENTS

Column
payment_id
booking_id
amount
payment_method

23. Data Model Relationships

The primary relationships are:



These relationships allow Power BI to filter and aggregate operational data efficiently.

24. Key Performance Indicators

The dashboard contains several operational KPIs.

24.1 Total Flights

Number of flight records in the analytical dataset.

Total Flights = COUNTROWS(FACT_FLIGHTS)

24.2 Total Airlines

Number of unique airlines.

Total Airlines = DISTINCTCOUNT(FACT_FLIGHTS[airline])

24.3 Total Routes

Number of unique source-destination routes.

Total Routes = DISTINCTCOUNT(FACT_FLIGHTS[route])

24.4 Average Flight Duration

Average calculated flight duration in minutes.

Average Flight Duration =
AVERAGE(FACT_FLIGHTS[duration_minutes])

24.5 Overnight Flights

Number of flights where the arrival date is later than the departure date.

24.6 Total Anomalies

Number of records that have a non-valid data quality flag.

24.7 Minimum Flight Duration

Shortest calculated flight duration.

24.8 Maximum Flight Duration

Longest calculated flight duration.

24.9 Anomaly Rate

Anomaly Rate =
Total Anomalies / Total Flights

25. Power BI Dashboard

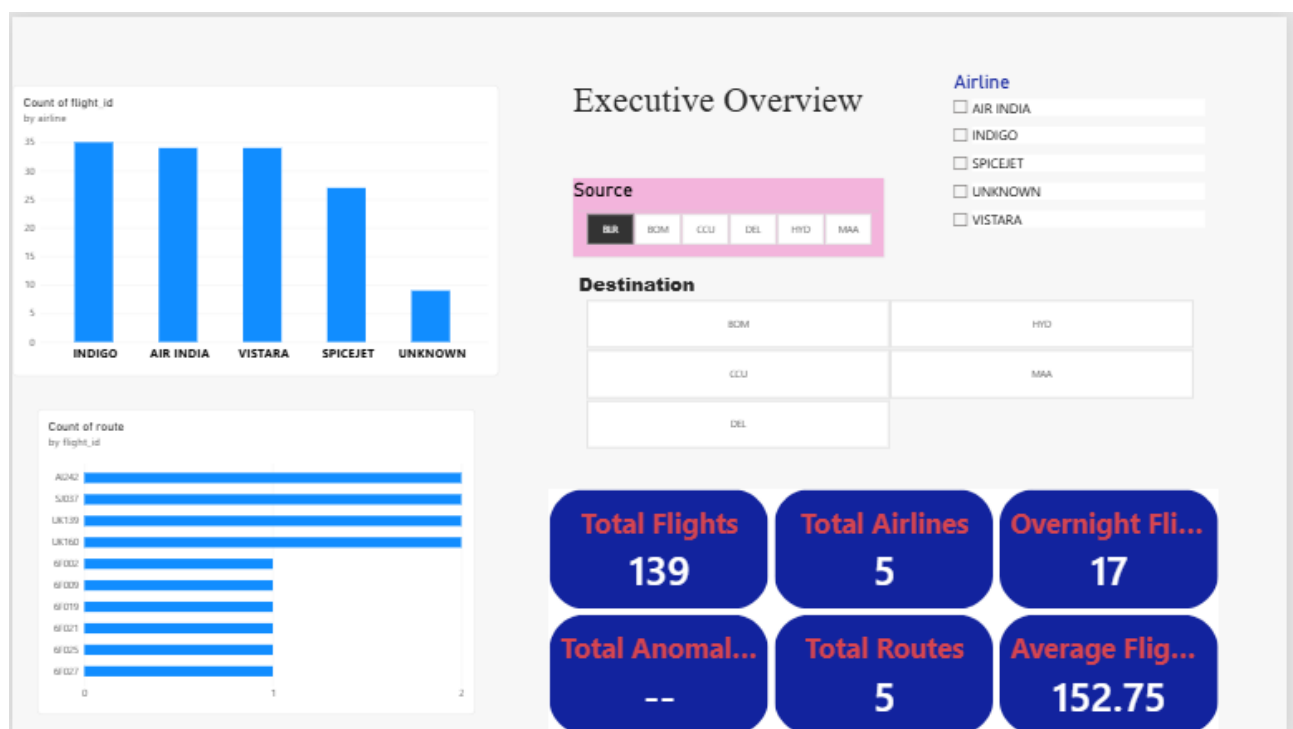
The Power BI report is designed to be interactive, business-friendly, and visually organized.

The dashboard is divided into multiple pages.

26. Dashboard Page 1 – Executive Overview

Page Title

ASG AIRLINES / Executive Overview



KPI Cards

1. Total Flights
2. Total Airlines
3. Total Routes
4. Average Flight Duration
5. Overnight Flights
6. Total Anomalies

Visualizations

Flight Distribution by Airline

A column chart showing the number of flights operated by each airline.

Top 10 Routes by Flight Count

A bar chart showing the busiest routes.

Average Flight Duration by Airline

A comparison of average flight durations between airlines.

Overnight vs Non-Overnight Flights

A donut chart showing the distribution.

Flights by Departure Day

A column chart showing flight volume across days of the week.

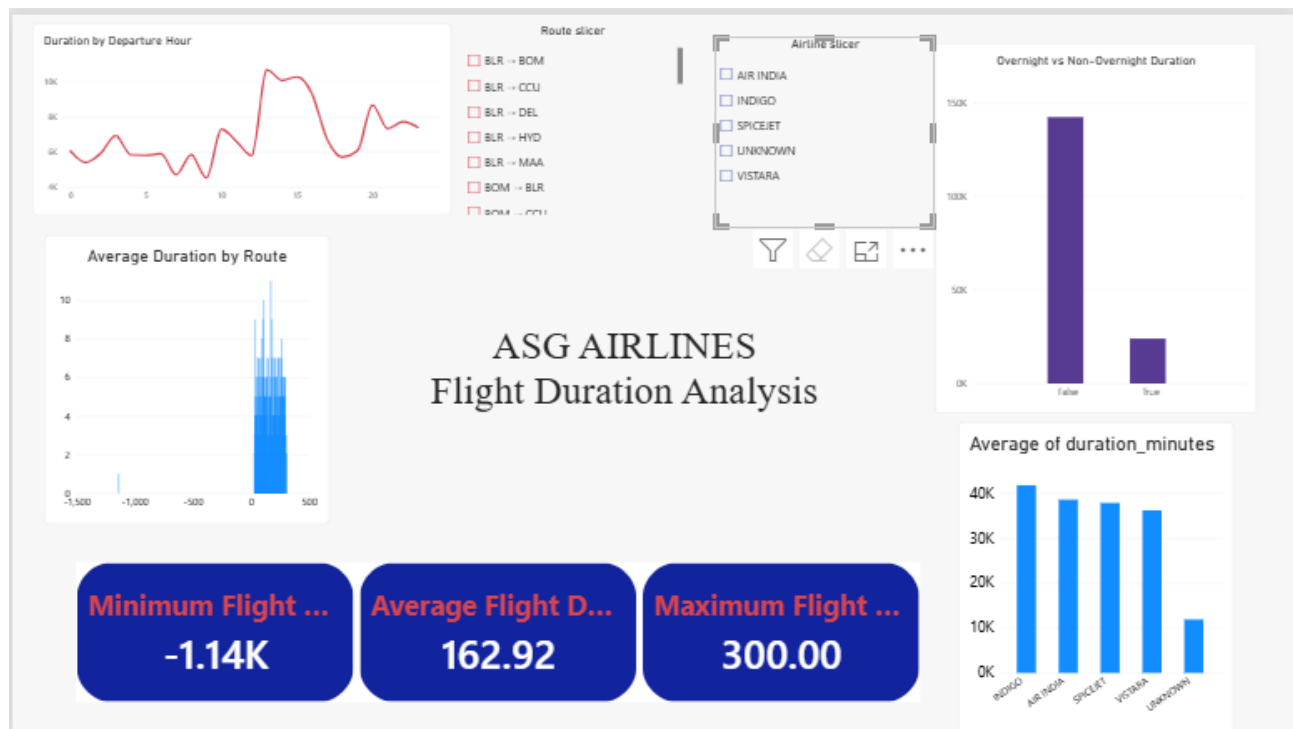
Slicers

- Airline
- Route
- Source Airport
- Destination Airport

27. Dashboard Page 2 – Duration Analysis

Page Title

ASG AIRLINES / Flight Duration Analysis



KPI Cards

- Average Duration
- Minimum Duration
- Maximum Duration

Visualizations

1. Average duration by airline
2. Average duration by route
3. Average duration by departure hour
4. Average duration for overnight and non-overnight flights

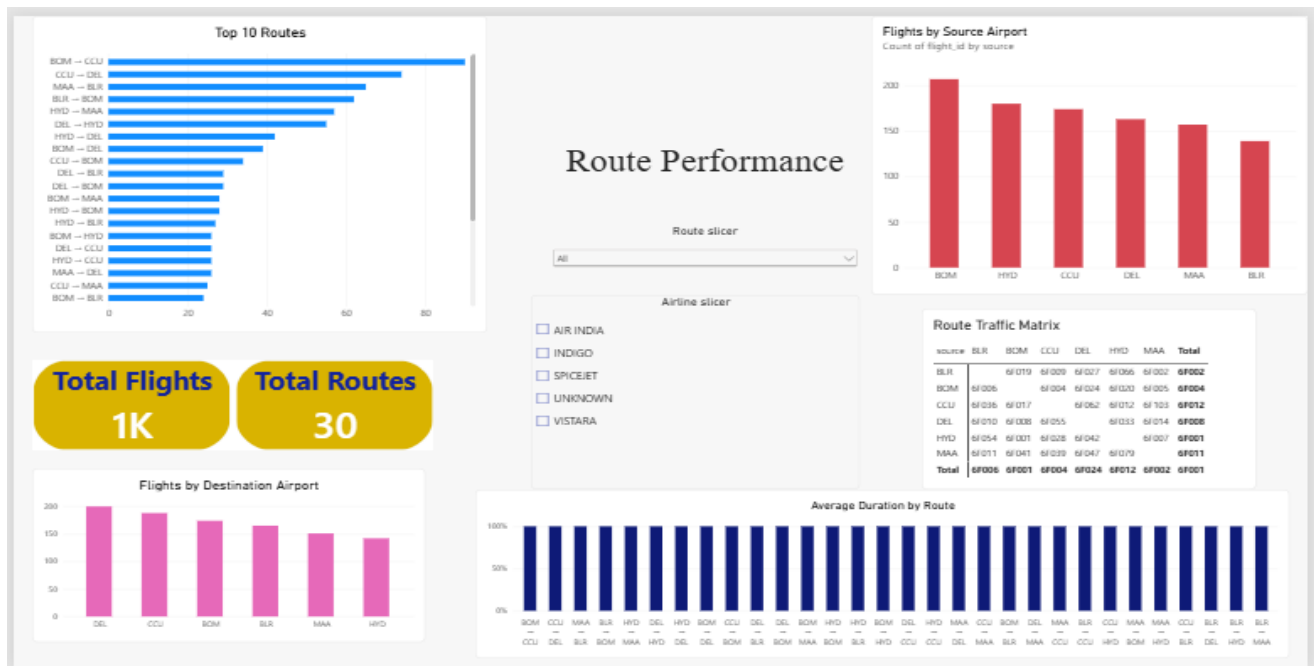
Slicers

- Airline
- Route

28. Dashboard Page 3 – Route Performance

Page Title

ASG AIRLINES / Route Performance



KPI Cards

- Total Flights
- Total Routes

Visualizations

1. Top 10 routes by traffic
2. Flights by source airport
3. Flights by destination airport
4. Average flight duration by route
5. Source-to-destination traffic matrix

Business Use

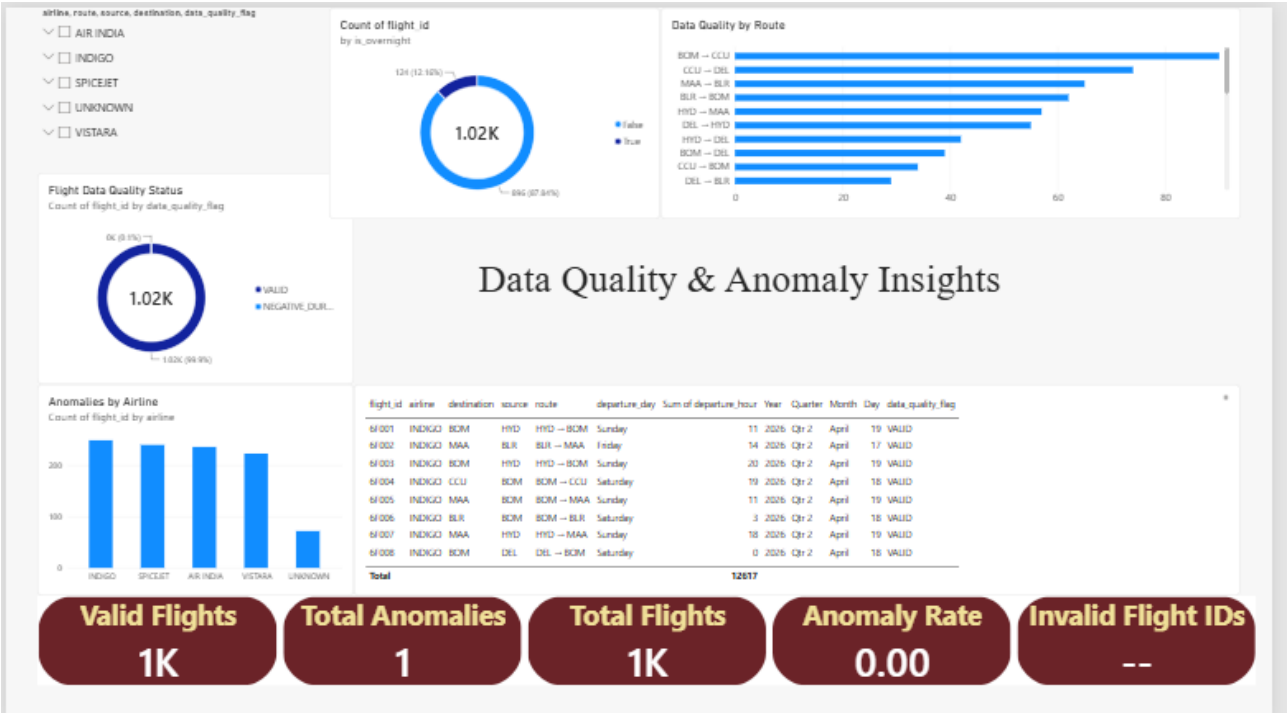
This page allows airline management to identify:

- High-volume routes
- Low-volume routes
- Important airports
- Routes with higher average duration

29. Dashboard Page 4 – Data Quality and Anomaly Insights

Page Title

ASG AIRLINES / Data Quality & Anomaly Insights



KPI Cards

- Total Flights
- Total Anomalies
- Valid Flights
- Anomaly Rate
- Invalid Flight IDs

Visualizations

Flight Data Quality Status

Donut chart showing:

VALID
INVALID

Anomalies by Airline

Identifies airlines associated with data-quality issues.

Anomalies by Route

Shows routes containing anomalous records.

Duration Difference

Shows differences between recorded and calculated flight duration.

Flight Data Quality Details

A detailed table containing:

- Flight ID
- Airline
- Route
- Duration
- Calculated Duration
- Duration Difference
- Overnight Status
- Data Quality Flag

30. Important Limitation Regarding Flight Delays

The original dataset contains departure and arrival timestamps, but it does not provide separate:

- Scheduled departure time
- Actual departure time
- Scheduled arrival time
- Actual arrival time

Therefore, true flight delay minutes cannot be calculated reliably.

For this reason, the project should **not claim to calculate actual operational delays**.

Instead, the dashboard focuses on:

- Data anomalies
- Duration inconsistencies
- Overnight operations
- Route performance
- Flight distribution

If scheduled and actual timestamps become available in a future dataset, delay KPIs can be added.

31. DAX Measures

The following measures can be implemented in Power BI.

```
Total Flights =  
COUNTROWS(fact_flights)
```

```

Total Airlines =
DISTINCTCOUNT(fact_flights[airline])

Total Routes =
DISTINCTCOUNT(fact_flights[route])

Average Flight Duration =
AVERAGE(fact_flights[duration_minutes])

Overnight Flights =
CALCULATE(
    COUNTROWS(fact_flights),
    fact_flights[is_overnight] = TRUE()
)

Total Anomalies =
CALCULATE(
    COUNTROWS(fact_flights),
    fact_flights[data_quality_flag] <> "VALID"
)

Valid Flights =
CALCULATE(
    COUNTROWS(fact_flights),
    fact_flights[data_quality_flag] = "VALID"
)

Minimum Flight Duration =
MIN(fact_flights[duration_minutes])

Maximum Flight Duration =
MAX(fact_flights[duration_minutes])

Anomaly Rate =
DIVIDE(
    [Total Anomalies],
    [Total Flights],
    0
)

```

32. Pipeline Implementation Workflow

The pipeline follows these steps:

Step 1 – Ingestion

Read the Excel workbook.

Step 2 – Profiling

Inspect:

- Shape
- Columns
- Data types

- Missing values
- Duplicate records
- Unique categorical values

Step 3 – Cleaning

Perform:

- Text standardization
- Missing-value handling
- Identifier validation
- Timestamp validation

Step 4 – Transformation

Generate:

- Flight duration
- Duration in minutes
- Duration in hours
- Overnight indicator
- Route
- Date features
- Time features

Step 5 – Data Governance

Apply:

- Passenger identifier hashing
- Aadhaar masking
- PII removal
- Public-data protection

Step 6 – Validation

Check:

- Referential integrity
- Negative durations
- Invalid timestamps
- Invalid flight IDs
- Missing required values

Step 7 – Analytical Modeling

Create:

- Fact tables
- Dimension tables

Step 8 – Export

Export cleaned and analytical datasets as CSV files.

Step 9 – Power BI

Import the analytical datasets and build interactive dashboards.

33. Error Handling and Logging

The pipeline should include appropriate error handling.

Examples of errors that should be handled include:

- Missing input file
- Missing worksheet
- Invalid column names
- Invalid timestamp values
- Incorrect data types
- Failed transformation
- Missing relationships
- Export failures

This improves troubleshooting and maintainability.

34. Scalability and Performance

The proposed architecture is designed with scalability in mind.

For the current dataset, Pandas is sufficient because the dataset is relatively small.

For significantly larger datasets, the following improvements can be introduced:

1. PySpark for distributed processing.
2. Azure Databricks for large-scale transformations.
3. Azure Data Lake for scalable storage.
4. Incremental data processing.
5. Partitioning by flight date.
6. Parquet instead of CSV for analytical storage.
7. Data pipeline orchestration using Azure Data Factory.
8. Power BI semantic models for reporting.

Partitioning flight data by date can improve query performance when analyzing specific time periods.

35. Validation Strategy

After transformation, the following validation checks are performed.

Structural Validation

Check that expected columns exist.

Data Type Validation

Ensure:

- IDs are strings
- Dates are datetime values
- Amounts are numeric
- Duration is numeric

Missing Value Validation

Review missing values after cleaning.

Duration Validation

Check that:

Calculated Duration ≥ 0

Overnight Validation

Check:

Arrival Date > Departure Date

for overnight flights.

Referential Integrity

Check:

Booking Flight ID $\rightarrow$ Flight ID
Booking Passenger ID $\rightarrow$ Passenger ID
Payment Booking ID $\rightarrow$ Booking ID

Duplicate Validation

Check primary identifiers for duplicate records.

36. Assumptions

The following assumptions were made during implementation.

1. `flight_id` is intended to uniquely identify a flight.
2. Source and destination represent airport codes.
3. Departure and arrival timestamps represent the actual flight event times available in the dataset.
4. Flight duration is calculated from arrival and departure timestamps rather than relying exclusively on the supplied duration.
5. A flight is classified as overnight when its arrival date is later than its departure date.
6. Missing airline values are represented as UNKNOWN where appropriate.
7. Invalid identifiers are flagged rather than automatically deleted.
8. Sensitive passenger information is not required for operational KPI analysis.
9. Passenger PII is masked, hashed, or excluded from public analytical datasets.

37. Enhancements Beyond Basic Requirements

The project includes several enhancements beyond the minimum requirements.

37.1 Data Quality Score

A future enhancement can calculate a percentage score based on:

$$\text{Valid Records} / \text{Total Records} \times 100$$

37.2 Duration Consistency Analysis

The difference between recorded and calculated duration can identify inconsistent operational records.

37.3 Overnight Flight Analytics

Flights crossing midnight are explicitly identified and analyzed.

37.4 Route Ranking

Routes can be ranked according to flight traffic.

37.5 Airline Performance

Airlines can be compared based on:

- Number of flights
- Average duration

- Overnight flights
- Data-quality issues

37.6 Interactive Filtering

Power BI slicers allow users to dynamically explore:

- Airlines
- Routes
- Source airports
- Destination airports
- Data quality status

38. Business Insights

The analytical solution enables airline management to answer questions such as:

Flight Operations

- How many flights are being operated?
- Which airlines operate the most flights?
- Which routes have the highest traffic?
- What is the average flight duration?

Route Analysis

- Which airports generate the highest traffic?
- Which routes are most frequently operated?
- Which routes have the longest average duration?

Overnight Operations

- How many flights operate overnight?
- Which airlines have more overnight operations?
- Which routes frequently cross midnight?

Data Quality

- How many flight records contain anomalies?
- Which airlines have more data-quality issues?
- Are recorded durations consistent with timestamp-derived durations?

39. Conclusion

The ASG Airlines data engineering project demonstrates a complete data pipeline from raw operational data to business intelligence reporting.

The solution addresses the major challenges present in the dataset, including:

- Missing values

- Inconsistent identifiers
- Timestamp standardization
- Flight duration calculation
- Overnight flight handling
- Referential integrity
- PII protection
- Data quality monitoring

The processed data is organized into analytical fact and dimension tables, making it suitable for Power BI reporting.

The Power BI dashboard provides operational insights through KPI cards, route analysis, airline analysis, duration analysis, overnight-flight analysis, and data-quality monitoring.

The architecture is also designed with future scalability in mind. While Python and Pandas are sufficient for the current dataset, and Power BI for production-scale airline data processing.